Chapter1: Computer System

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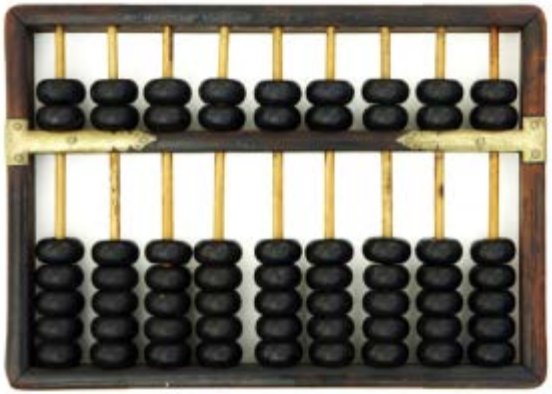
Learning Outcome

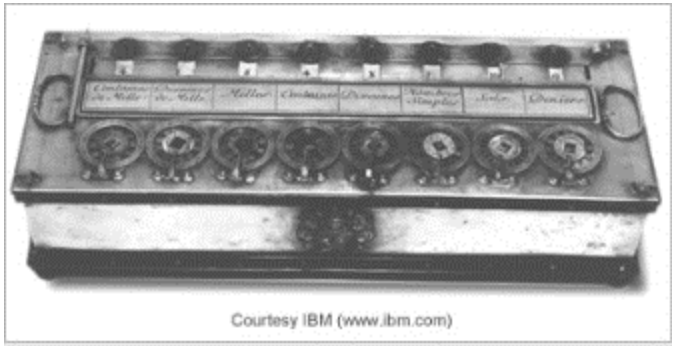
- Express computer evolution from the mechanical to semiconductor.
- Describe the difference between the microprocessor and microcontroller.

FIRST COMPUTING DEVICE

Mechanical computer



- Abacus
  - Used in China, Europa and Russia



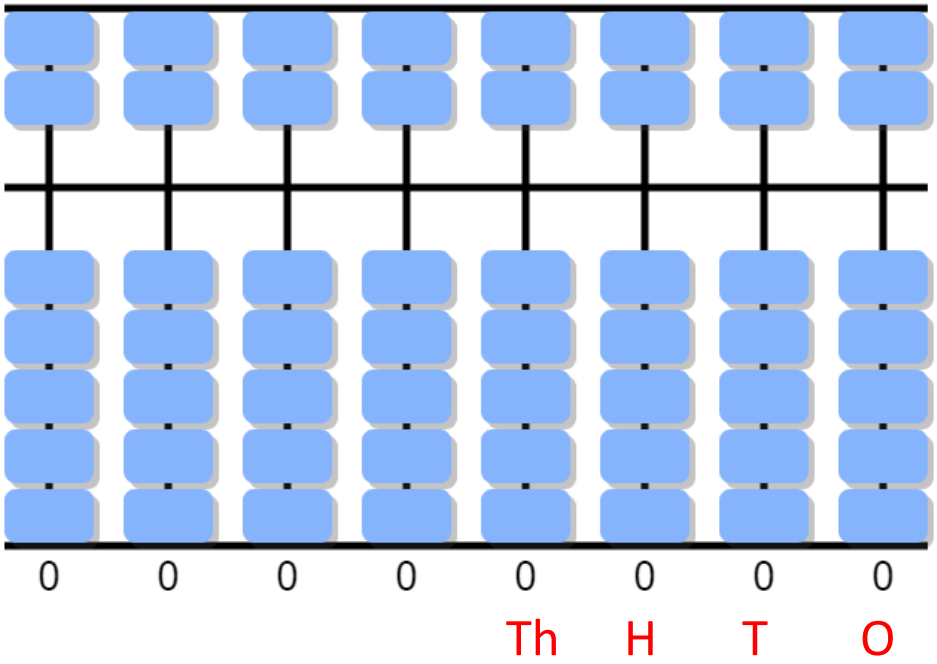
The Pascaline

- Mechanical calculator
  - Blaise Pascal (1642) developed a machine to help his father working in the shop.
  - Working only function of the addition and subtraction.

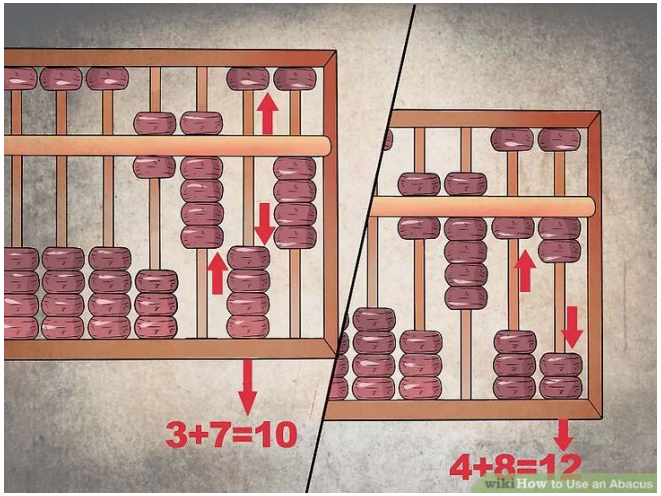
Activity 1.1 Play abacus

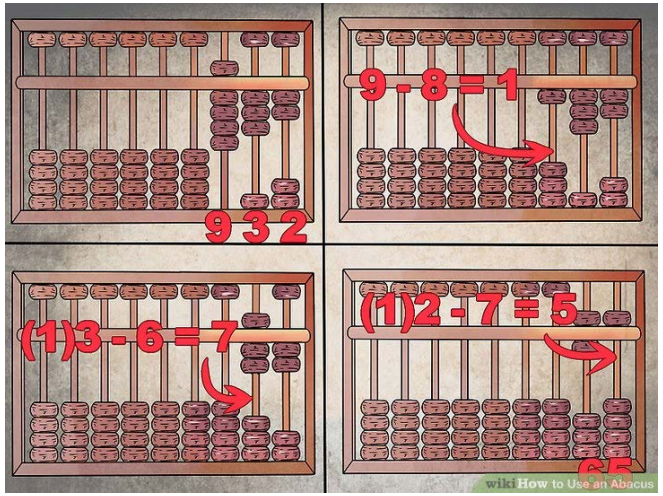
- Click on https://www.mathematik.uni-marburg.de/~thormae/lectures/ti1/code/abacus/sanpan.html

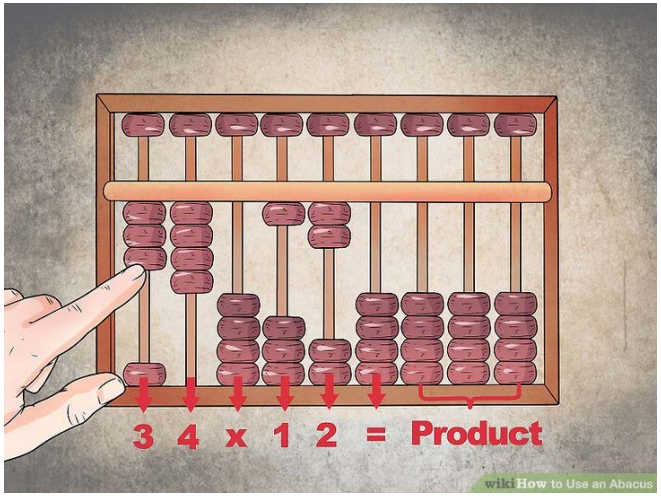
- Calculate
  - 83 + 12 = ?
  - 112 + 12 = ?

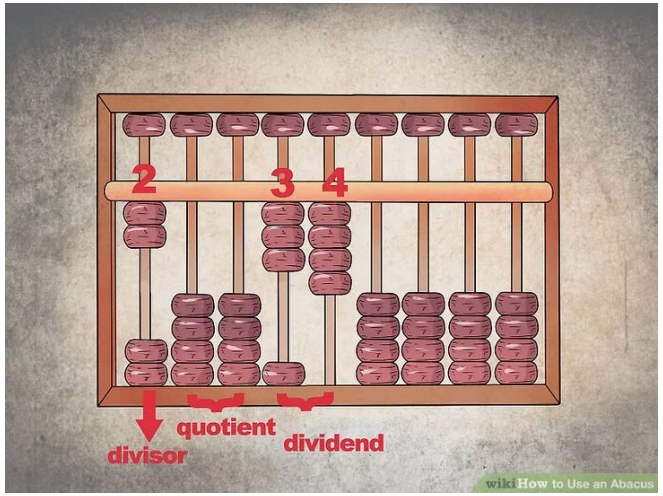


More functions in Abacus



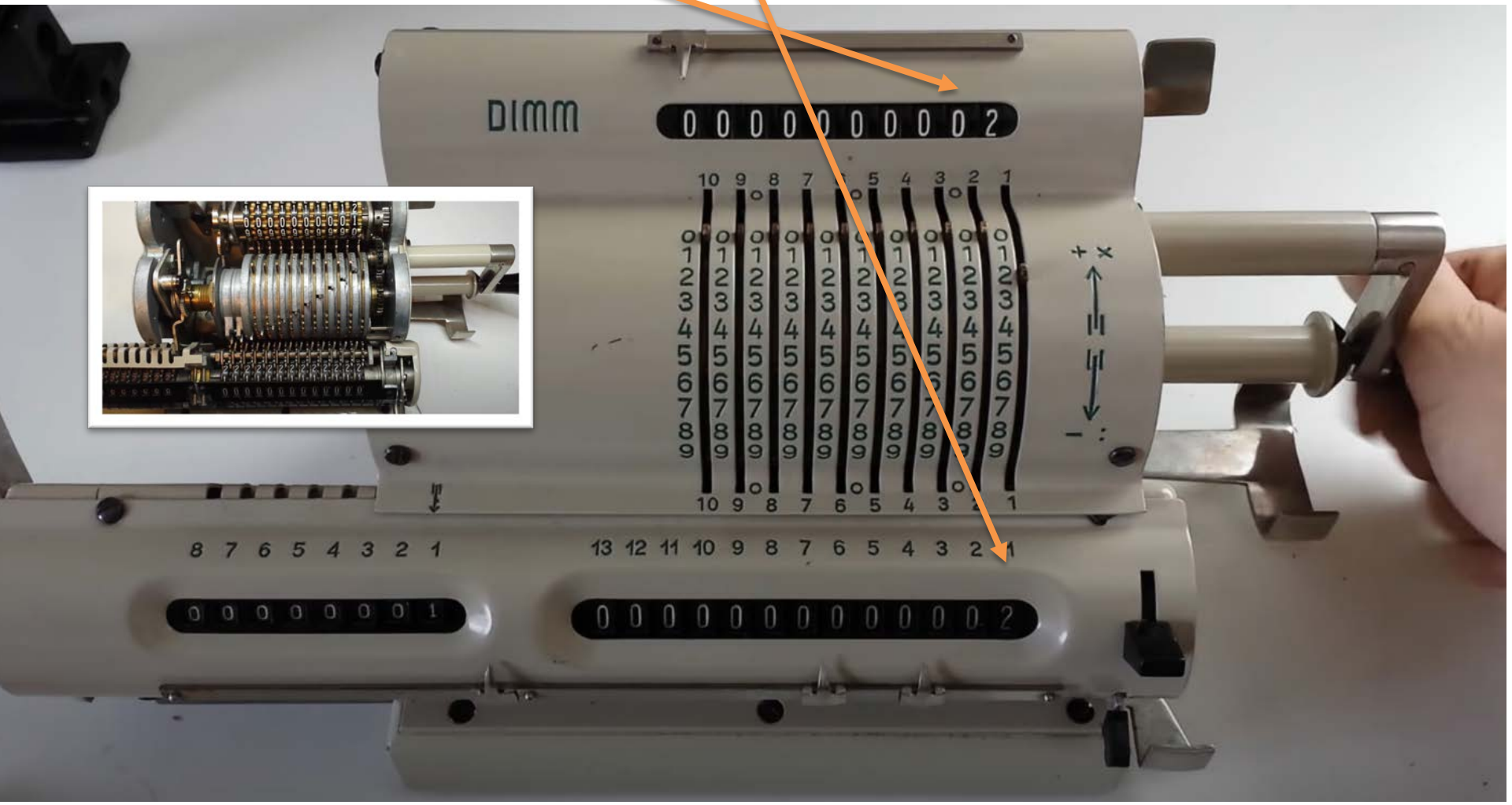






https://www.wikihow.com/Use-an-Abacus

2+2 = 4

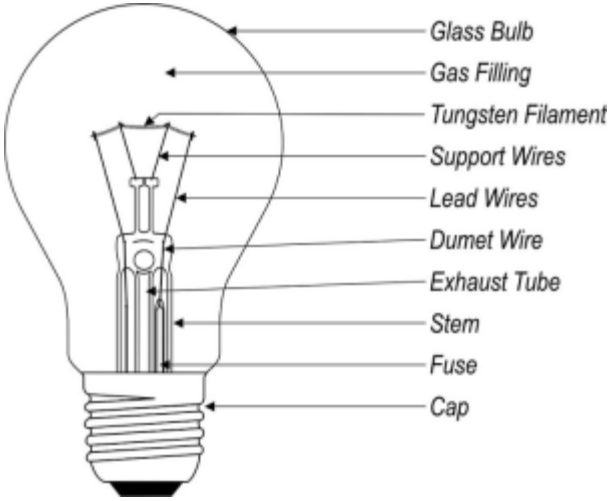


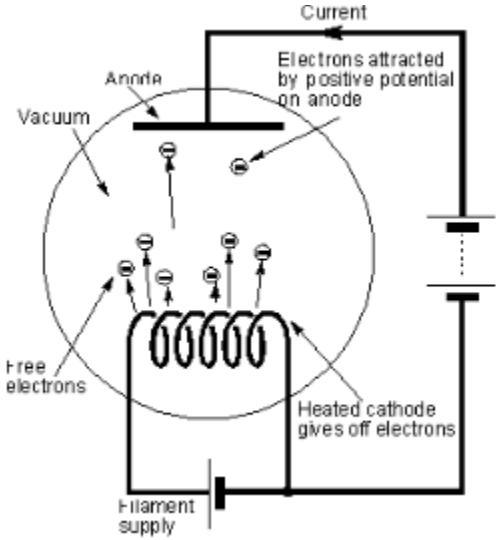
https://youtu.be/aDN4s8ElxqE?t=144

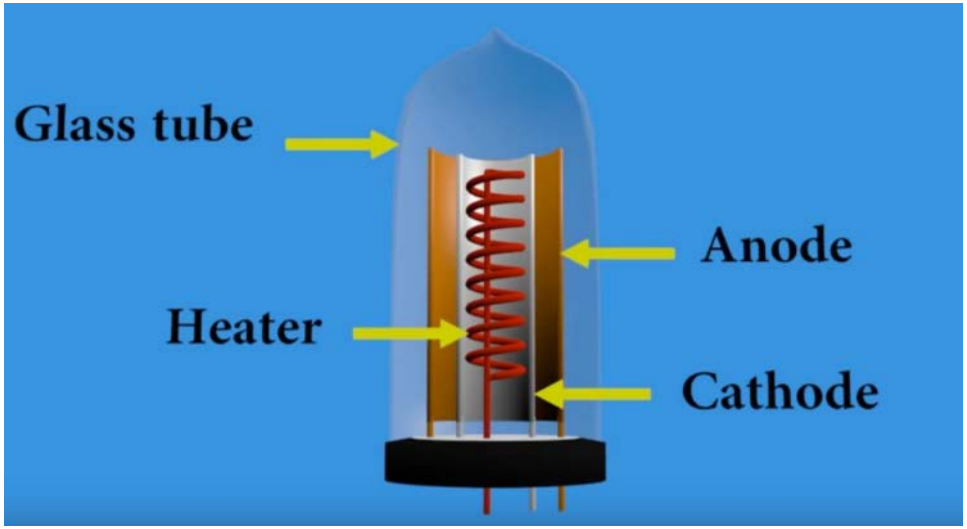
ELECTRIC AND ELECTRON

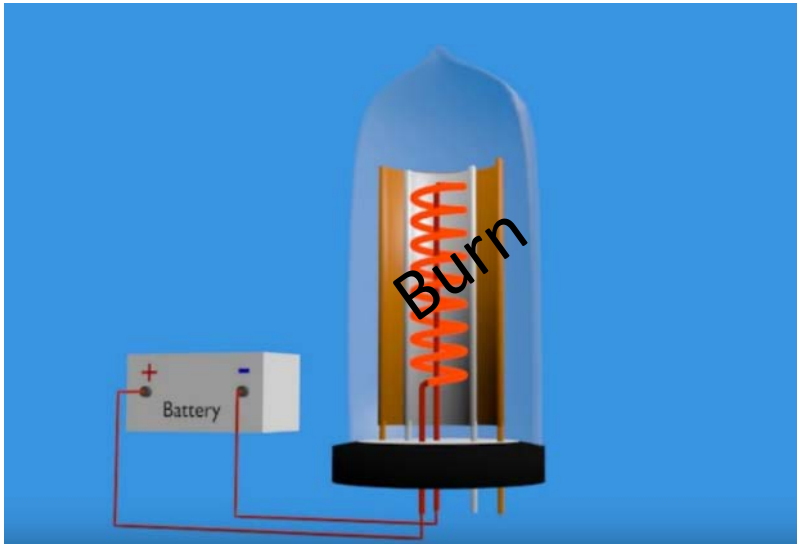
1.1.2 Vacuum Tube

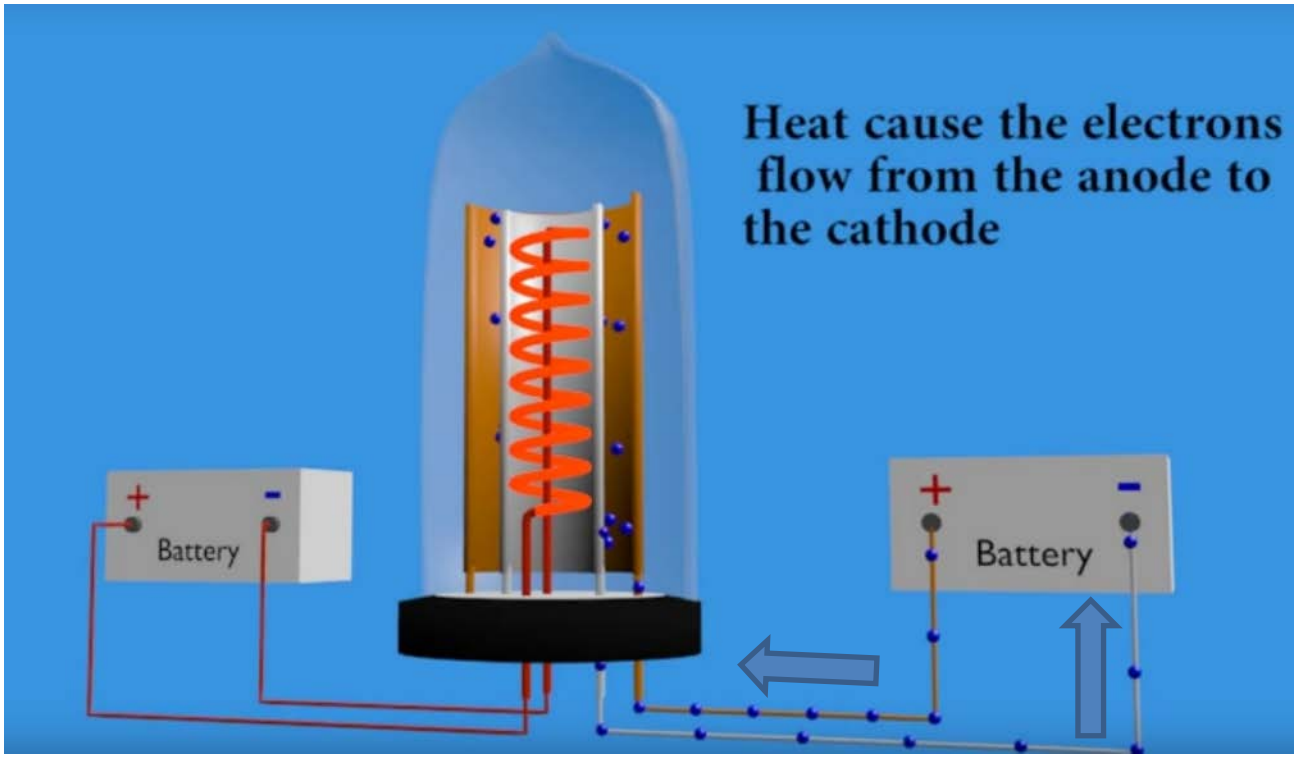
- Vacuum Tube was developed from *Incandescent lamp*.
- When *a tungsten*[74] *lead* gets heat, it spreads electron by moving to an anode plate.
- A grid plate uses controls number of electron moving from the tungsten lead to an anode plate.





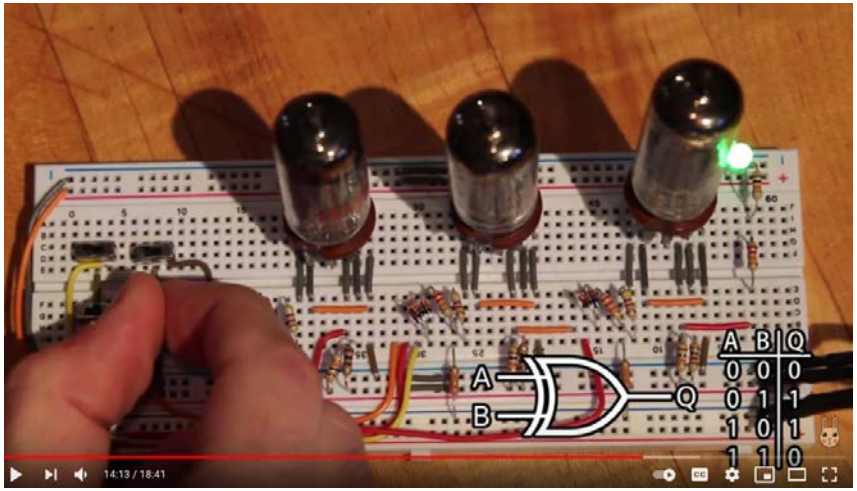






https://www.youtube.com/watch?v=K6BgZ8s1Vuw

Vacuum tube and future

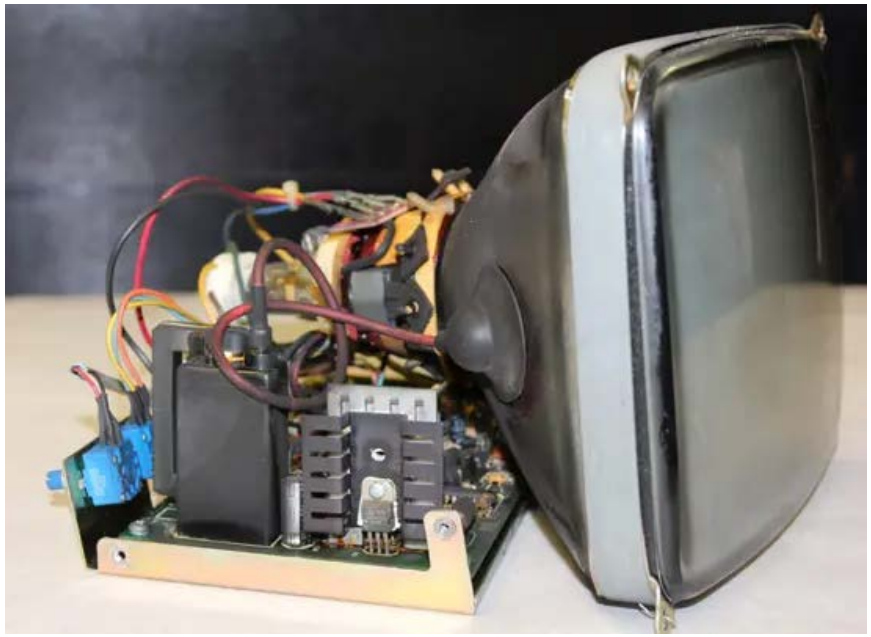


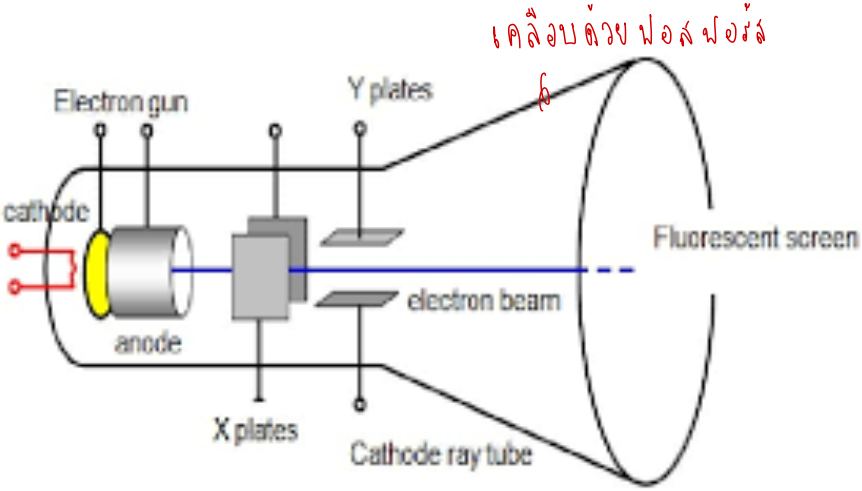
https://youtu.be/N-Sc6k_rlTM?t=853

4.) Future: The Nano Vacuum Tube

Vacuum tubes may make a comeback and replace standard microchips. Engineers have been able to build a structure in phosphorus doped silicon and use nanotubes to build a switch. These devices can operate 10 times faster than silicon transistors.

Cathod Ray Tube (CRT)

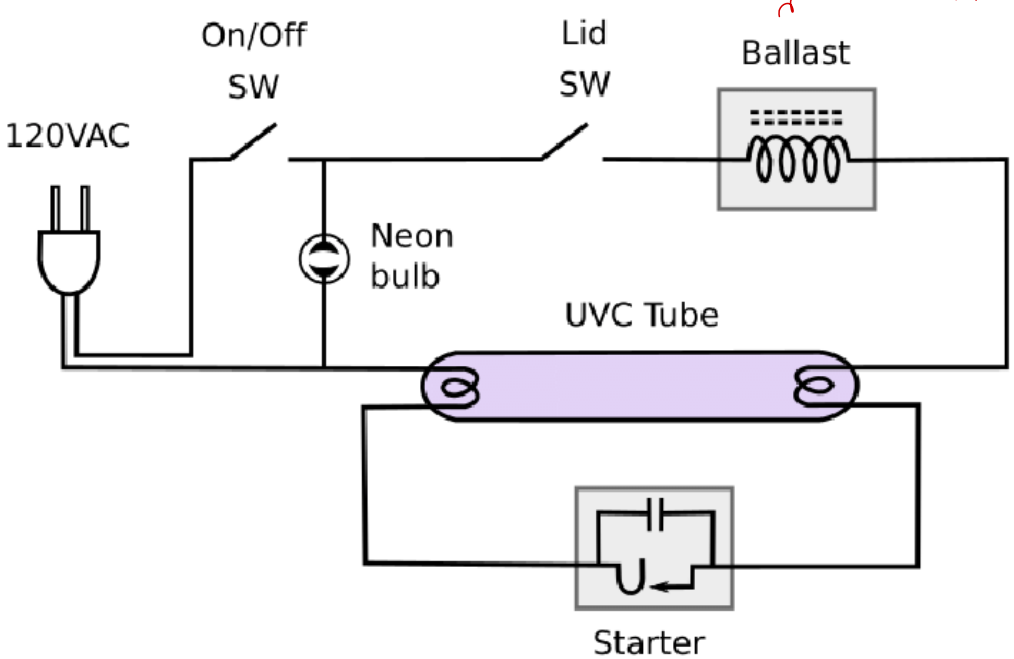




Activity1.2

- The fluorescence lamp has two small tungsten coils. Do you know why does the fluorescence lamp generates light? (Discussion)

ช่วยเพิ่มกระแสไฟฟ้า



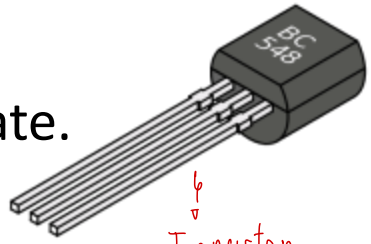
สายเคลือบไว้ต่อไฟฟ้าแรงสูง

SILICON CHIP

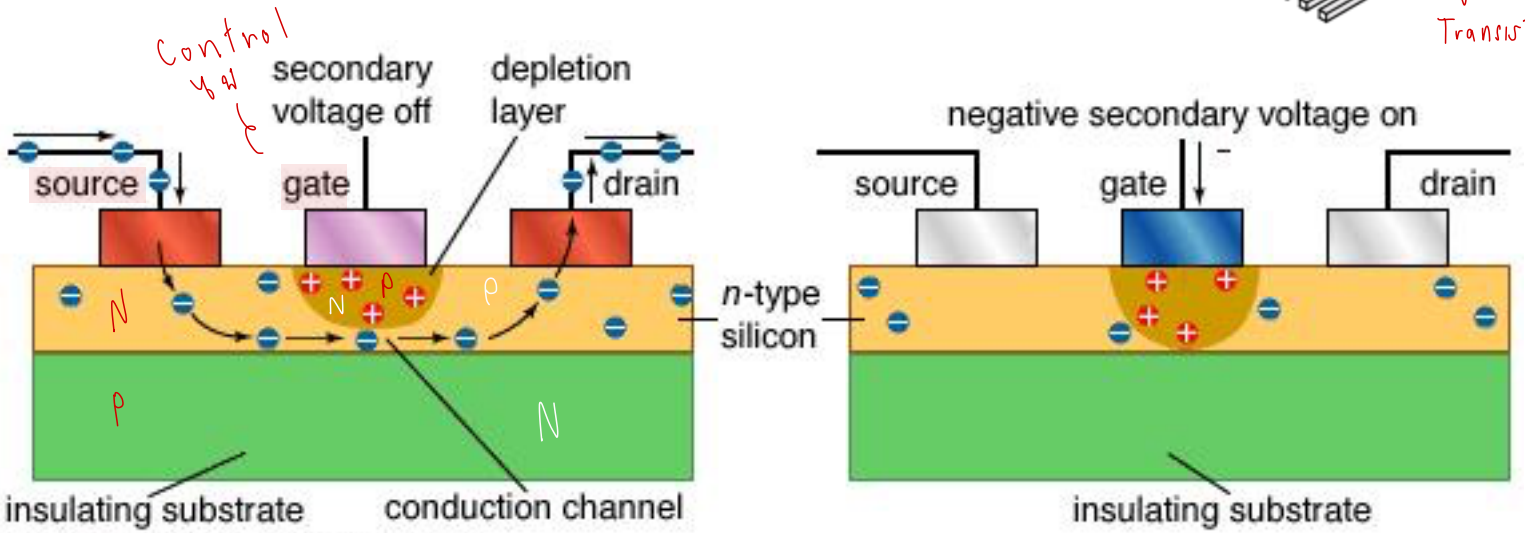
chip transister 90m ตัว
ใน core i7 gen 12

Silicon Transistor

- Semiconductor is created from Silicon[14] and Germanium[32].
- Electron movement from a source pin to a drain pin is controlled by a gate pin.
- SGD pins are installed in a silicon or germanium substrate.



Transister



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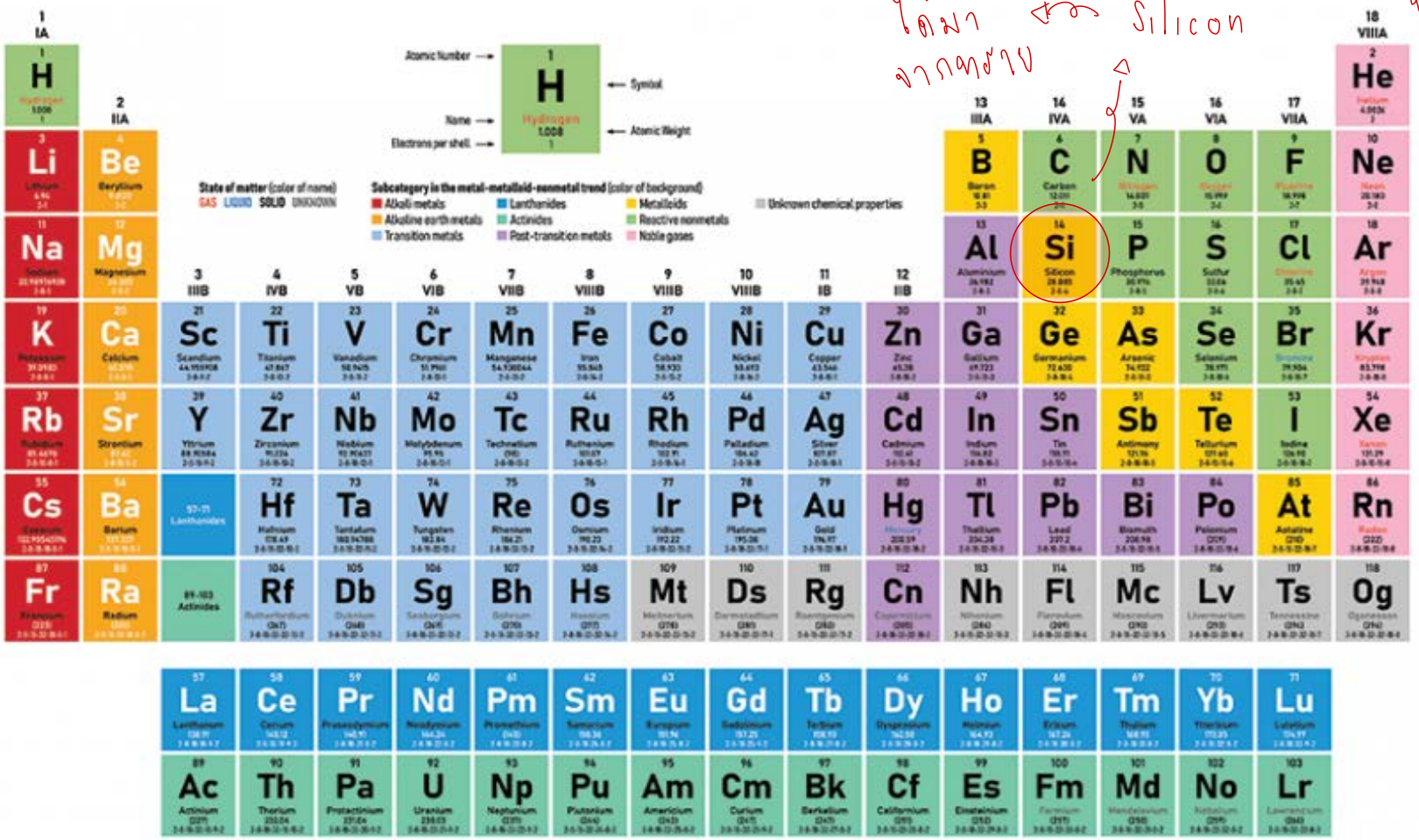
1.1.3 Silicon Transistor

- Where are Silicon, Germanium, Boron, and Antimony on the periodic table?

electron ที่ ความไวต่อไฟฟ้าสถิต น้อยกว่า

ได้มา จากซ้าย Silicon



1.1.3 Silicon chip-making process

- Silicon
  - Make from the ordinary sand, quartz, rock crystal, amethyst, agate, flint, jasper, and opal.

1.1.3 Silicon chip-making process

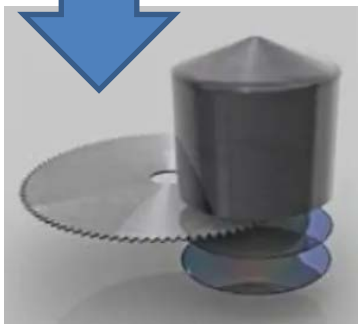
เผาเอาออกซิเจน
+ Oxygen / อัดไฟ

Burn

set

Pure silicon

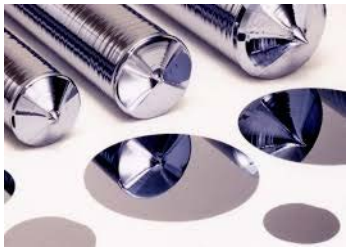
Cut



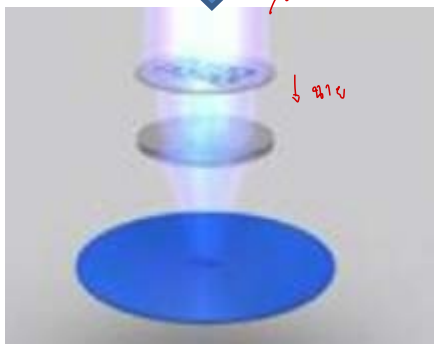
Wafer

ยังไม่สามารถทำอะไรได้

Exposure



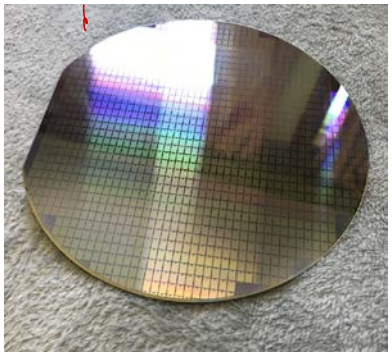
Engineer design circuit



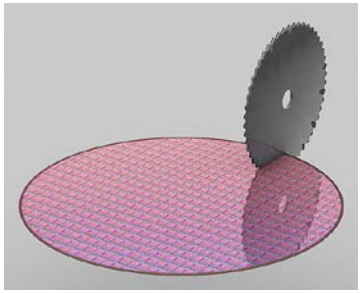
1.1.3 Silicon chip-making process

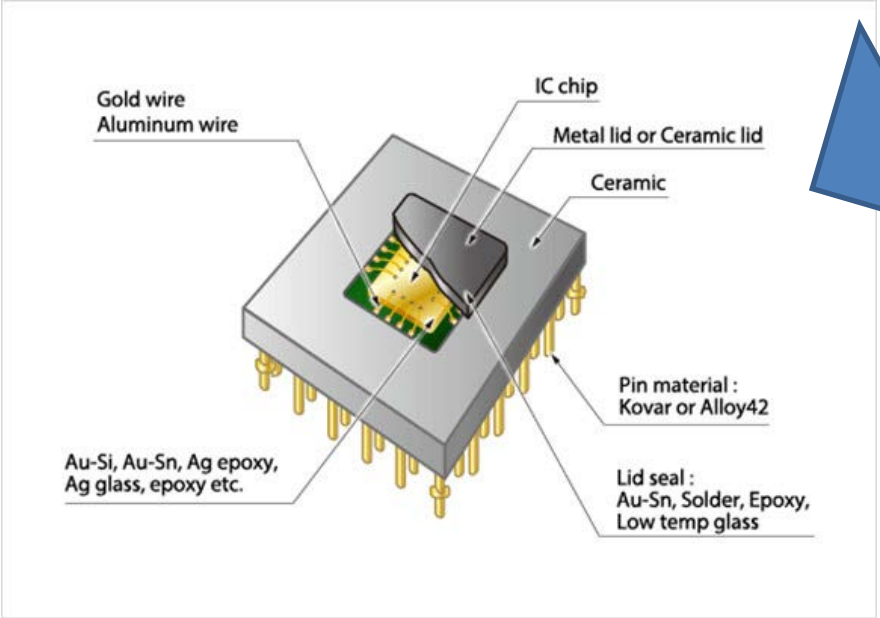
Expose
build each layer

Silicon 1 แผ่น
ø 12 inches



Cut and place
to package





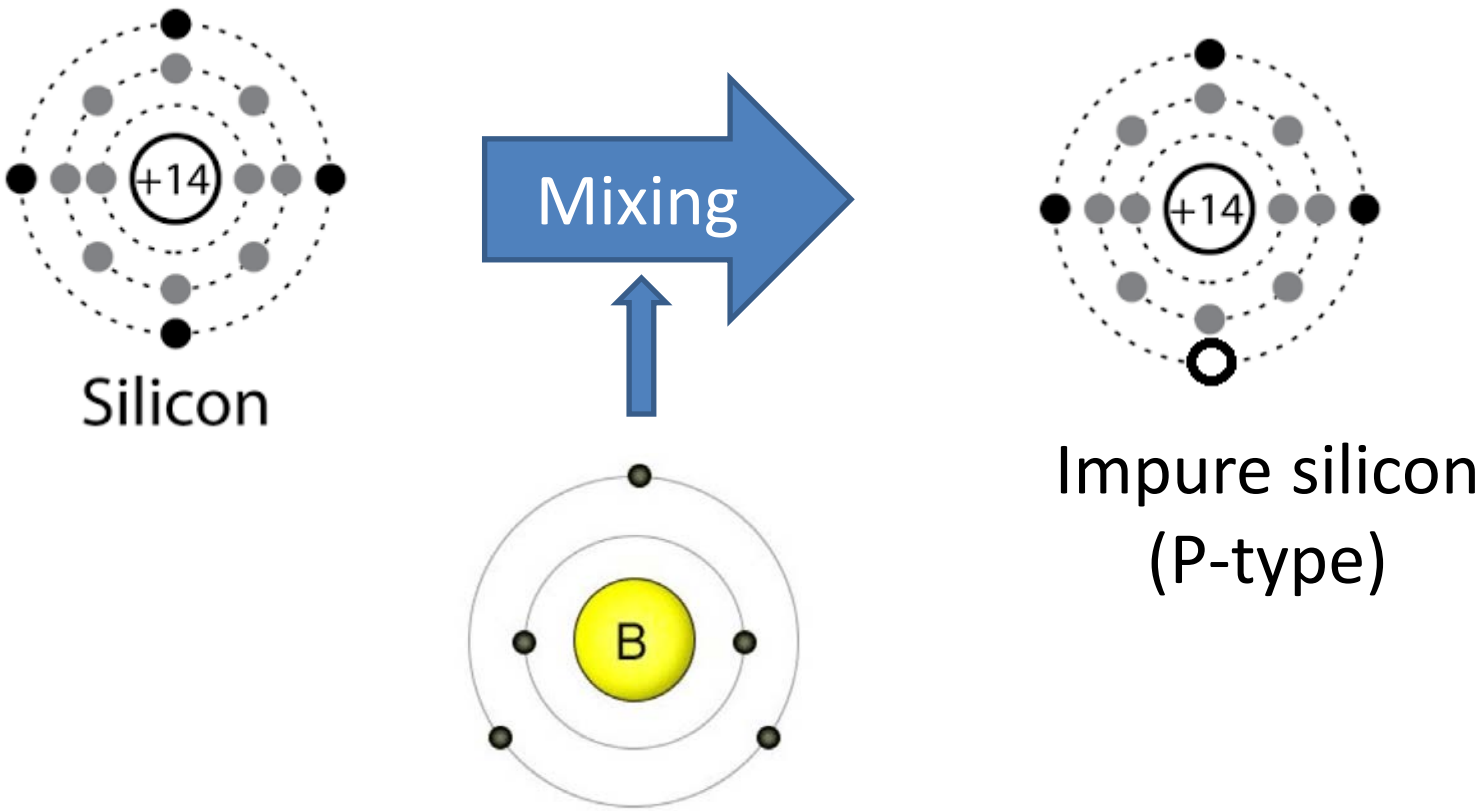
เอาไป check

ดี → ขาย premium

ไม่ดี → ขายราคาถูก

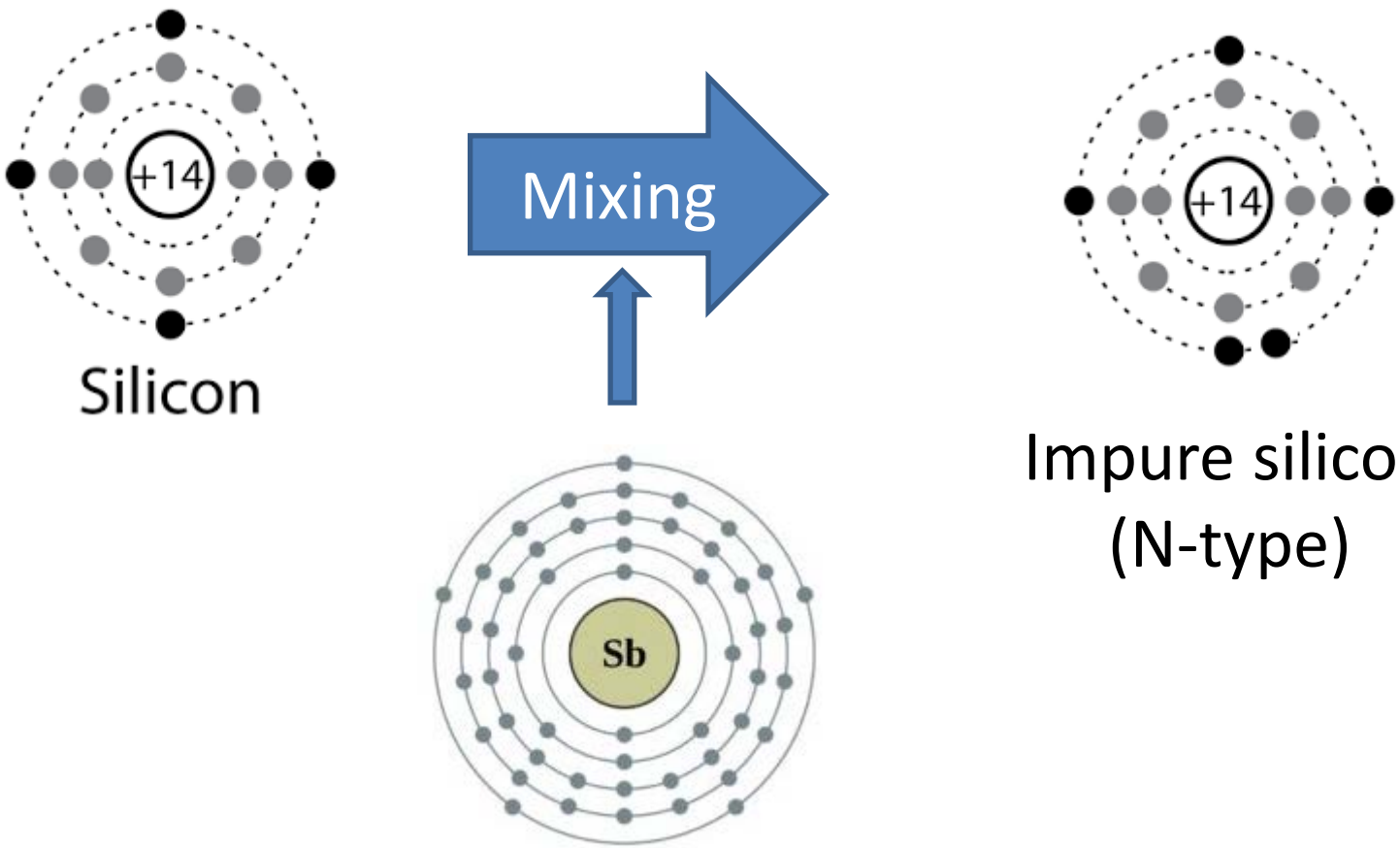
The main point of silicon chip

- Change from pure silicon to impure silicon by mixing boron to be P-type

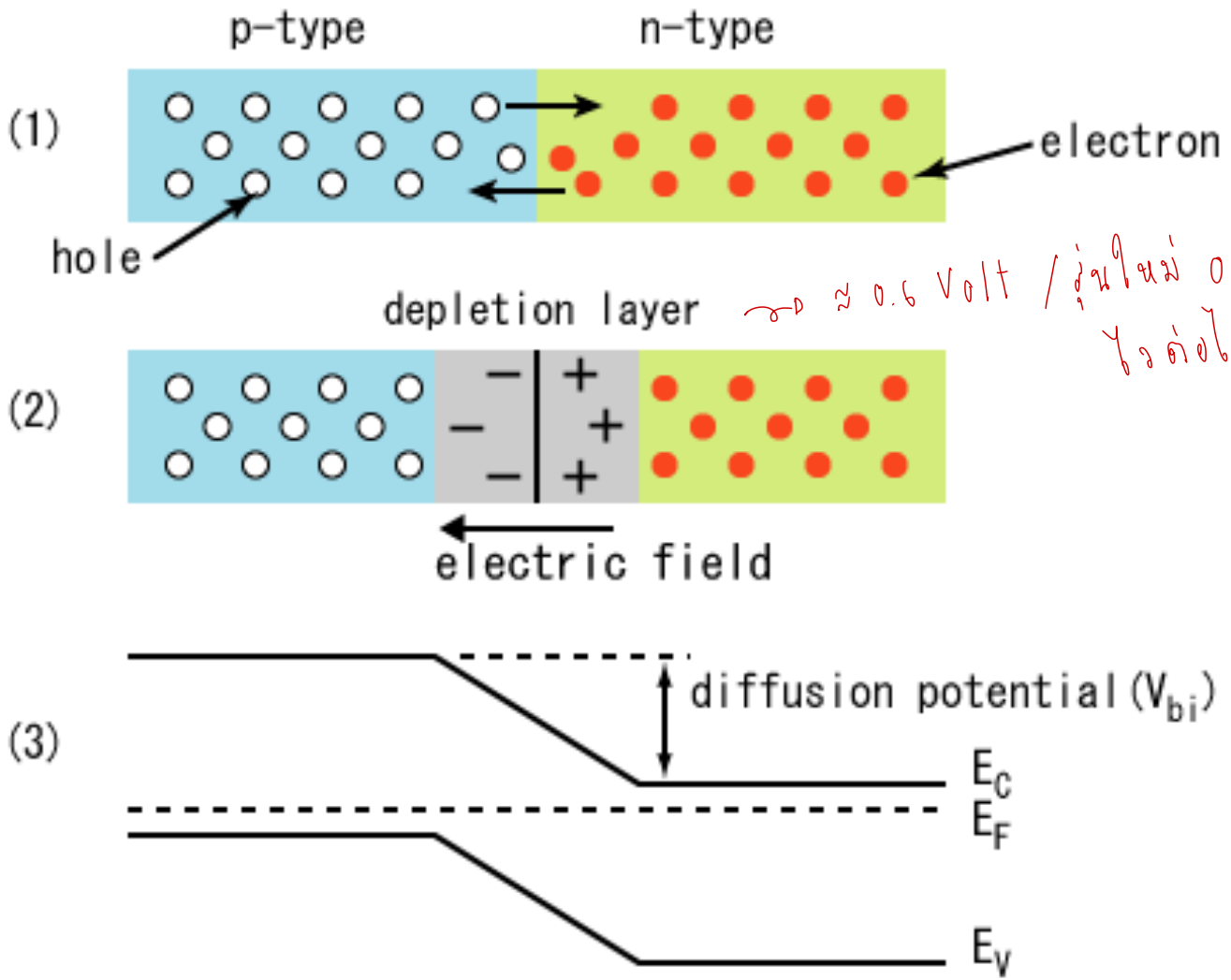


The main point of silicon chip

- Change from pure silicon to impure silicon by mixing antimonyแร่พวง to be N-type



N-type and P-type

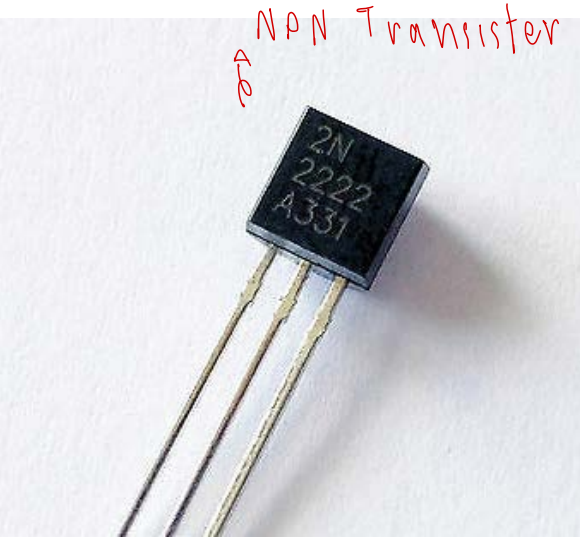


ซิลิ ≈ 0.6 Volt / ซุ่นใหม่ 0.3 Volt
ใช้ต่อไฟฟ้าสถิตมาก

Activity1.3

- Find information or specification of IC or Transistor from Internet.

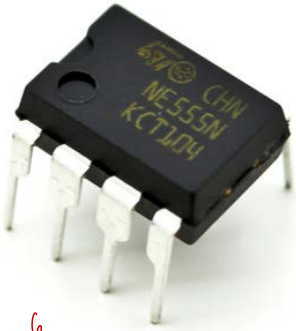
NPN Transister



IC

วงจรรวม
มี Transister มาต่อ
กันหลายตัว

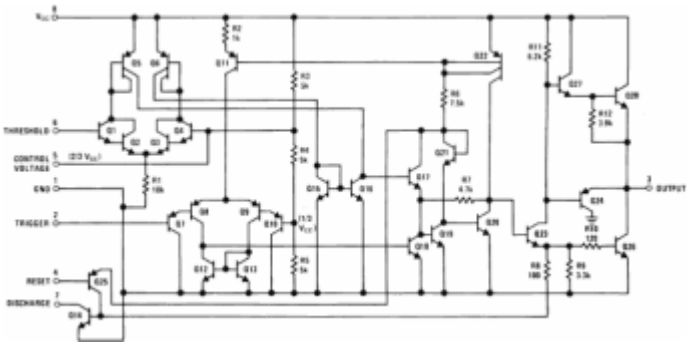
IC 555



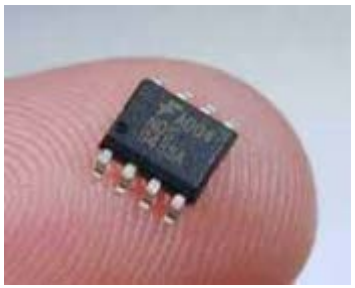
สร้างสัญญาณนาฬิกาได้

Integrated Circuit

- Discrete Circuit
- Integrated Circuit



Transister
20 กว่าตัว



IC

Integrated Circuit: Chip Scale

Name	Number of Transistors
Small Scale IC (SSI)	10
Medium SI (MSI)	10 – 1000
Large SI (LSI)	1K – 100K
Very LSI (VLSI)	100K – 1M
Ultra LSI (ULS)	> 1M

ขนาดทรานซิสเตอร์

Year	Size of the transistor in chip
1971 - 1985	10 µm – 1 µm
1989 - 1999	800nm – 180nm
2001 - 2010	130nm – 32nm
2012 - 2017	22nm – 10nm
App. 2018	7nm
App. 2020	5nm

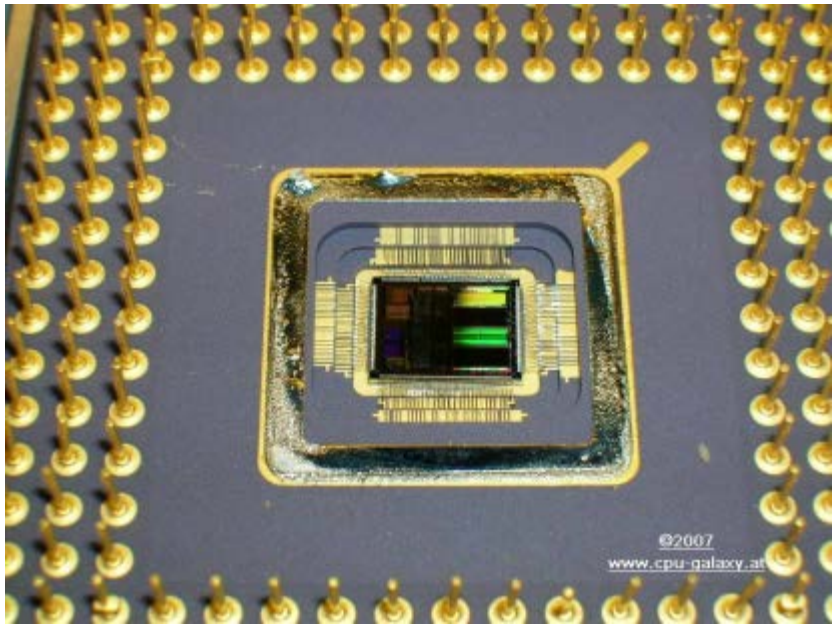
ถ้าเส้นของกมาเป็น 5000 m ส่วน ใน 1 m

What is the effectiveness when developing small size of the chip?

CPU die size

Years	Marketing names	Fabrication process
1971	Intel 4004 cpu 4 bit	10um
1982	Intel 80286	1.5um
1994	Intel 80486	180nm
2004	Pentium IV (Prescott)	90nm
2006	Pentium IV (Cedar Mill)	65nm
2007	Core 2, Dual-core	45nm
2010	Core i3-i7	32nm
2012	Corei3-i7 (ivy bridge)	22nm
≈2014		14nm
≈2016		10nm
≈2018	core i7 / iPhone 12 →	7nm
≈2020		5nm

Chip

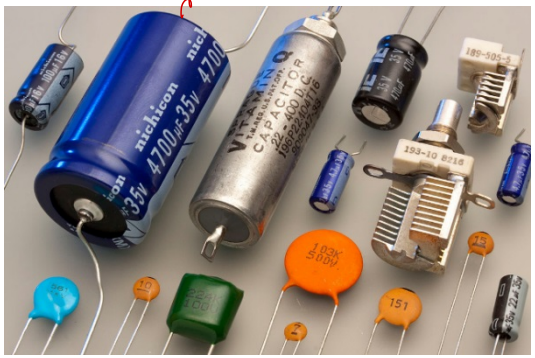


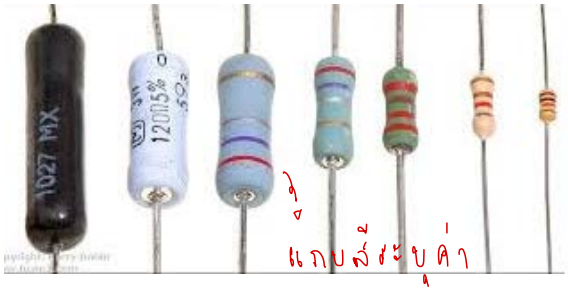
ELECTRONIC DEVICES

Electronic devices

ตัวกาง ทำให้คอมาเสีย
ไม่ทนความร้อน
(ด้านในเป็น Gel)
ชนิด Electrolize

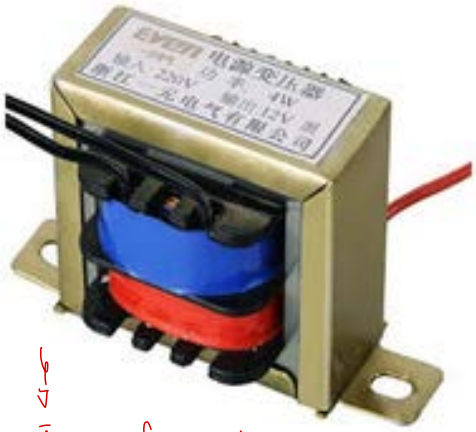
- Passive device
  - Capacitor ตัวเก็บประจุ
  - Resistor ตัวต้านทาน
  - Inductor ปล่อยความถี่ตามทิศทาง
  - Transformer



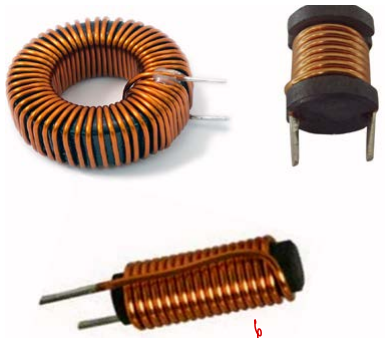


แถบสีระบุค่า

ความถี่สูงแกน
เป็น Carbon



Transformer

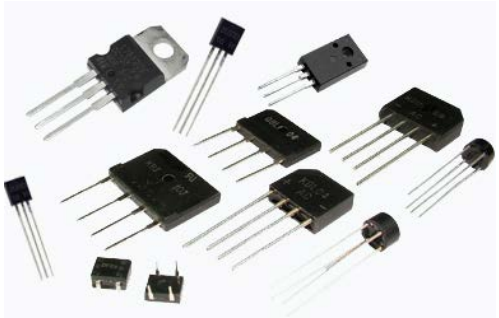


Inductor

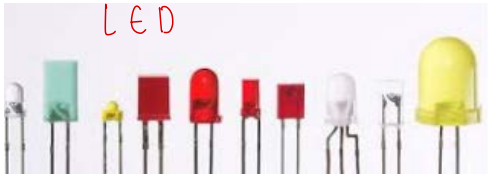
Electronic devices

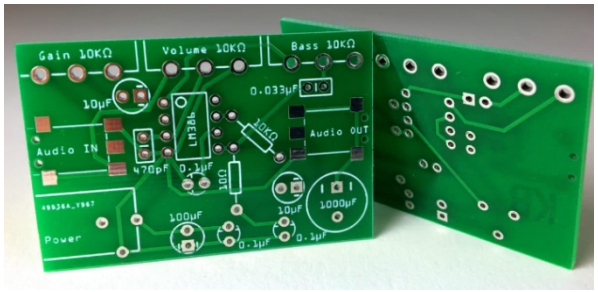
- Active device
  - Diode
  - Transistor
  - Light Emitting Diode (LED)
  - IC, Chip
- Component
  - Print Circuit Board (PCB)

Diode



LED





TYPE OF PROCESSOR

└ ชนิดของโพรเซสเซอร์

Computer Inside

- Processor
- Main memory
- System bus
- I/O Module

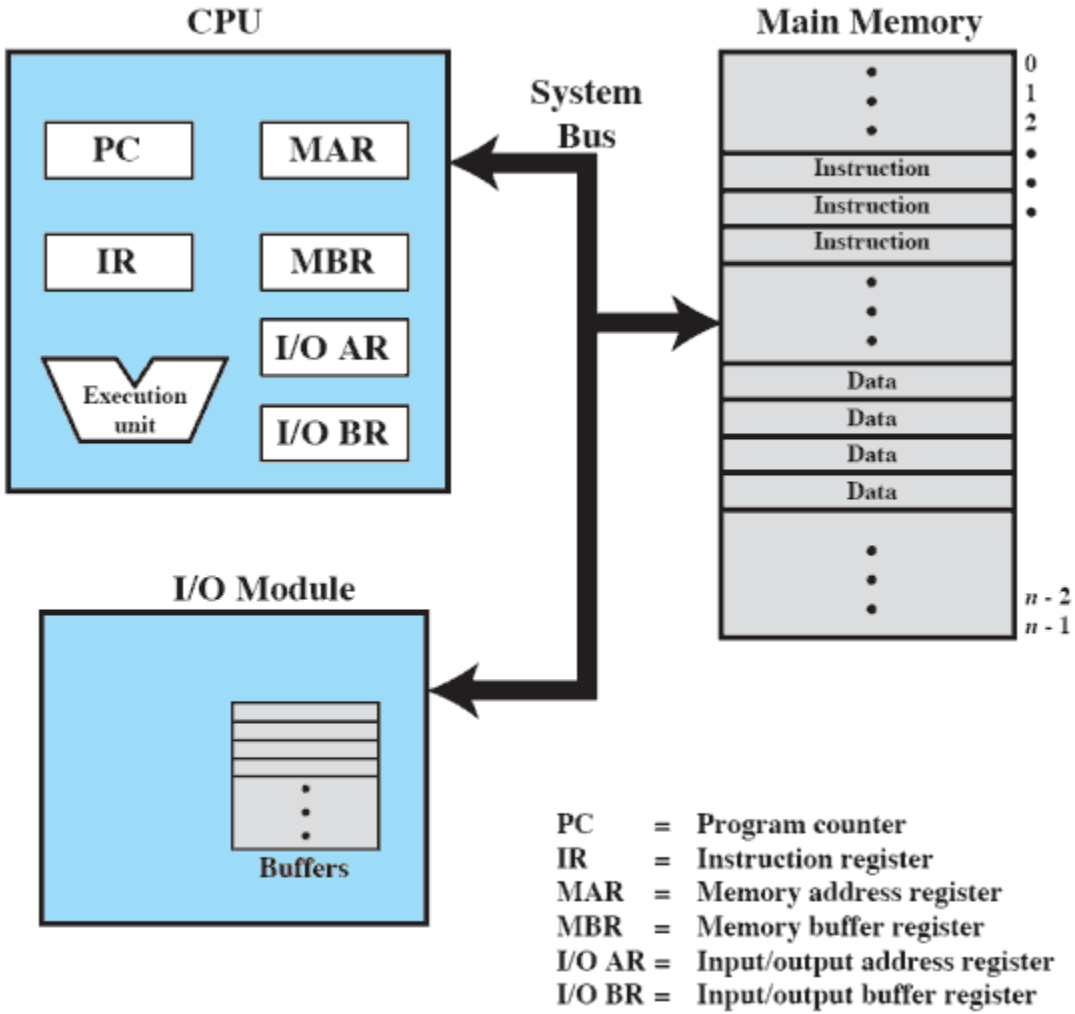
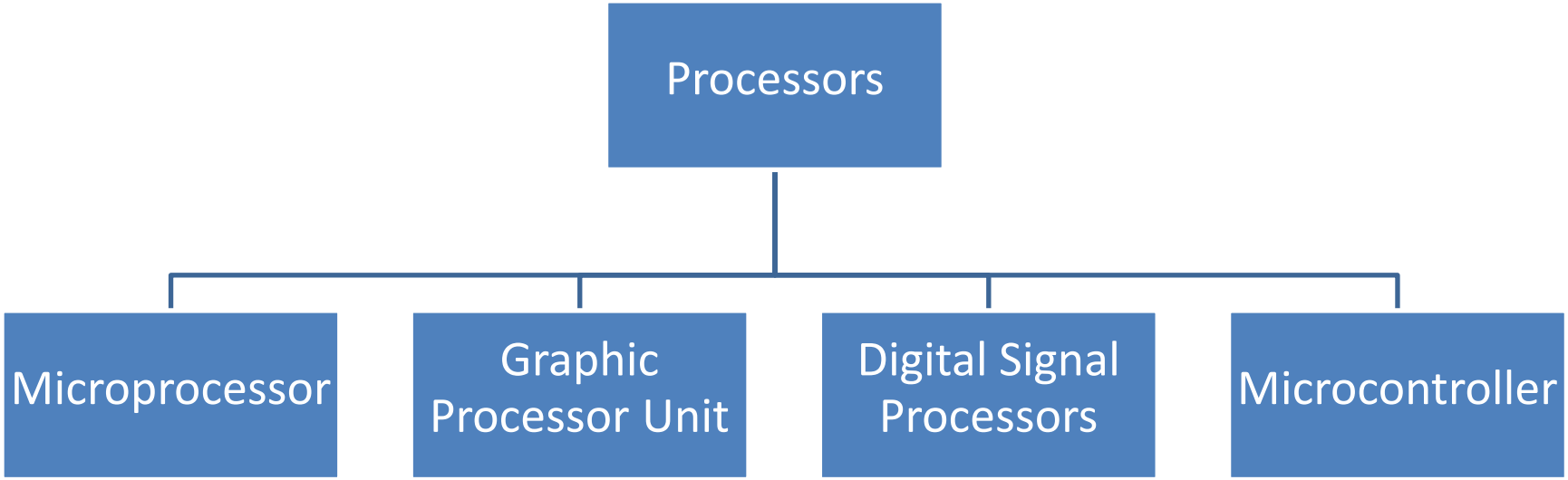
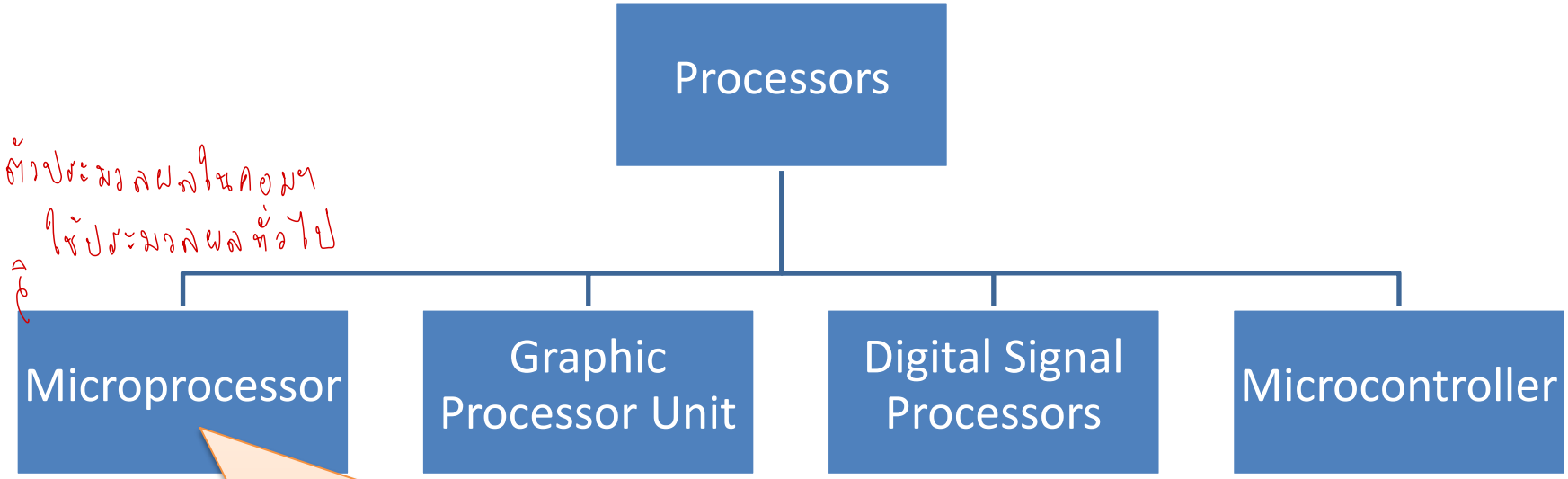
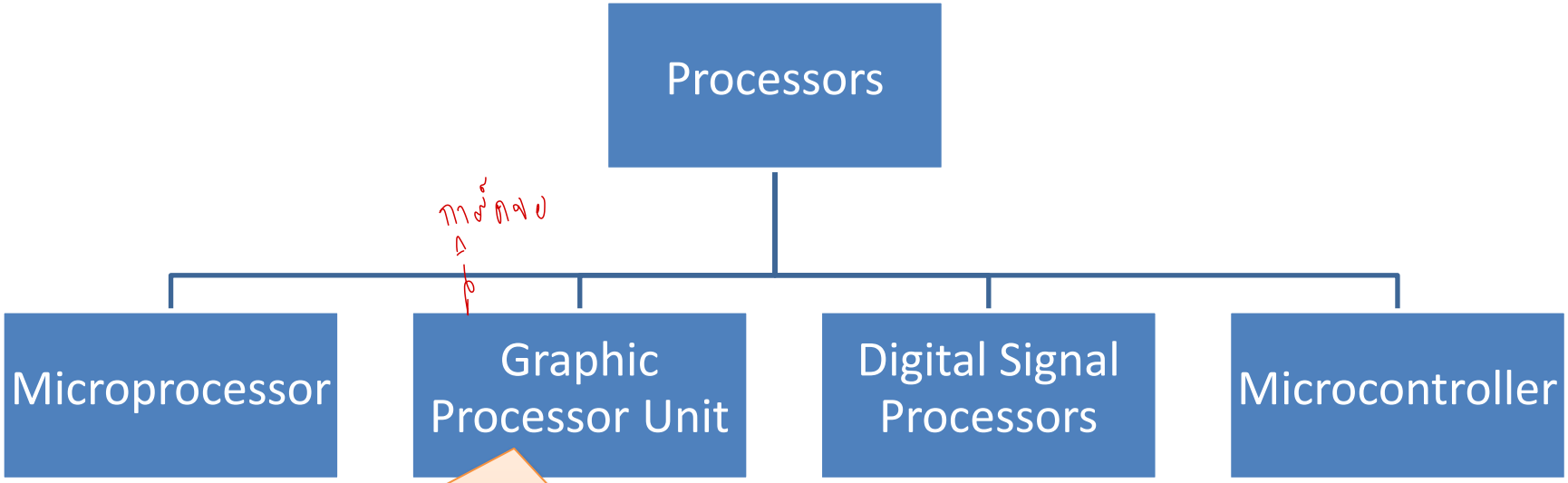


Figure 1.1 Computer Components: Top-Level View


Processors
Microprocessor
Graphic Processor Unit
Digital Signal Processors
Microcontroller



- General instruction set for general works
- No I/O ports directly
- Example
  - x86 CPUs
  - x64 CPUs



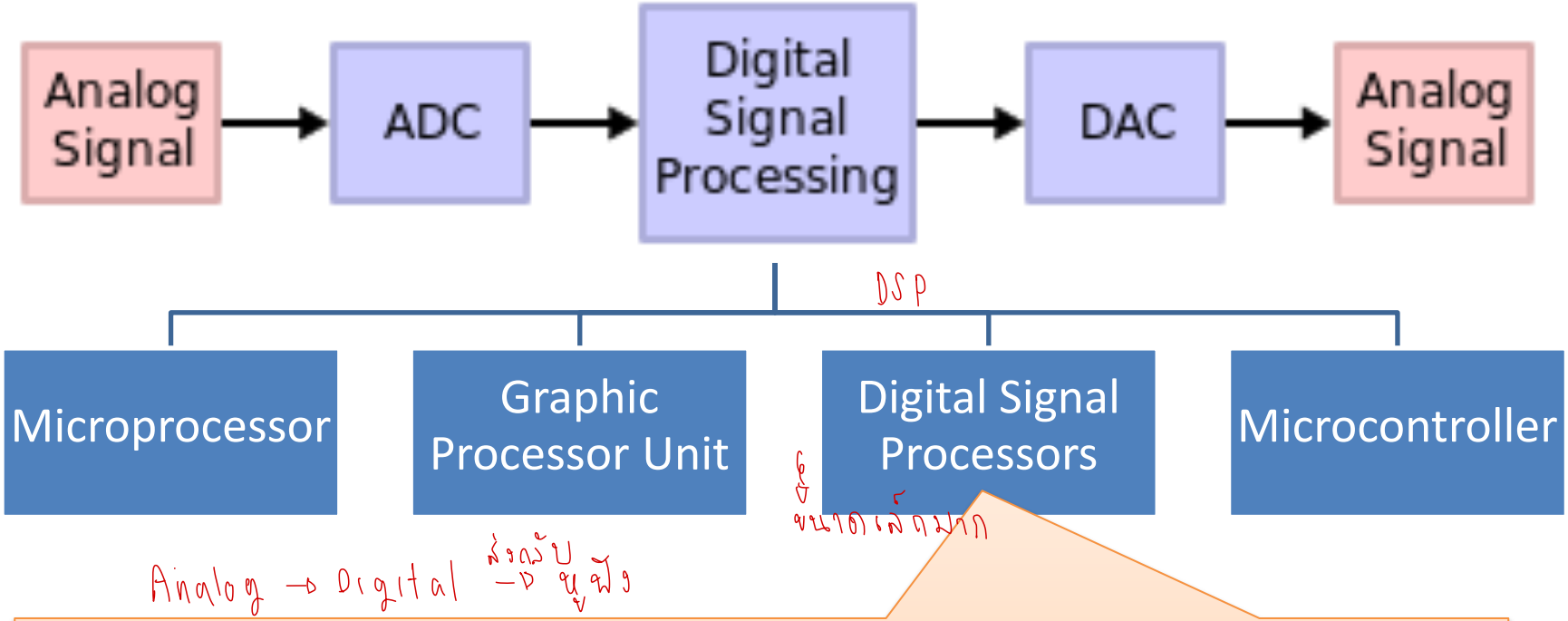
การ์ดจอ

Processor ที่มีคำสั่งเฉพาะสำหรับ Graphic

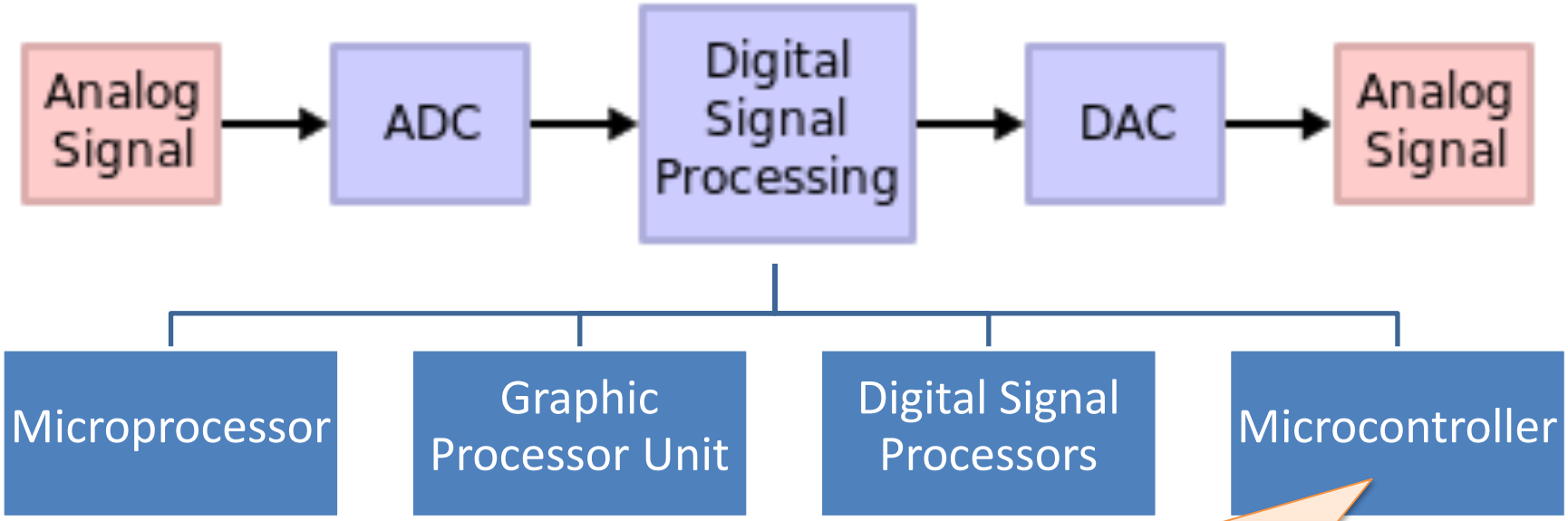
- Specific instruction set for vector and graphic computing and video rendering
- Dual ports memory interface
- Multicores
- Example family chips
  - GeForce, Radeon
  - Quadro, FirePro

การคำนวณบน Memory คำนวณเป็น Vector

Port



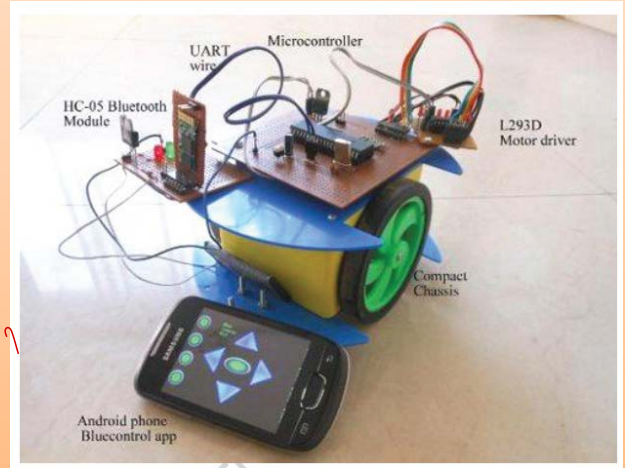
- Specific instruction set for streaming signal
  - Encoding audio/video
  - Decoding audio/video
- Example family chips
  - C6000 (Texas Instrument)
  - SHARC (Analog Devices)
  - EMU10K (Creative)



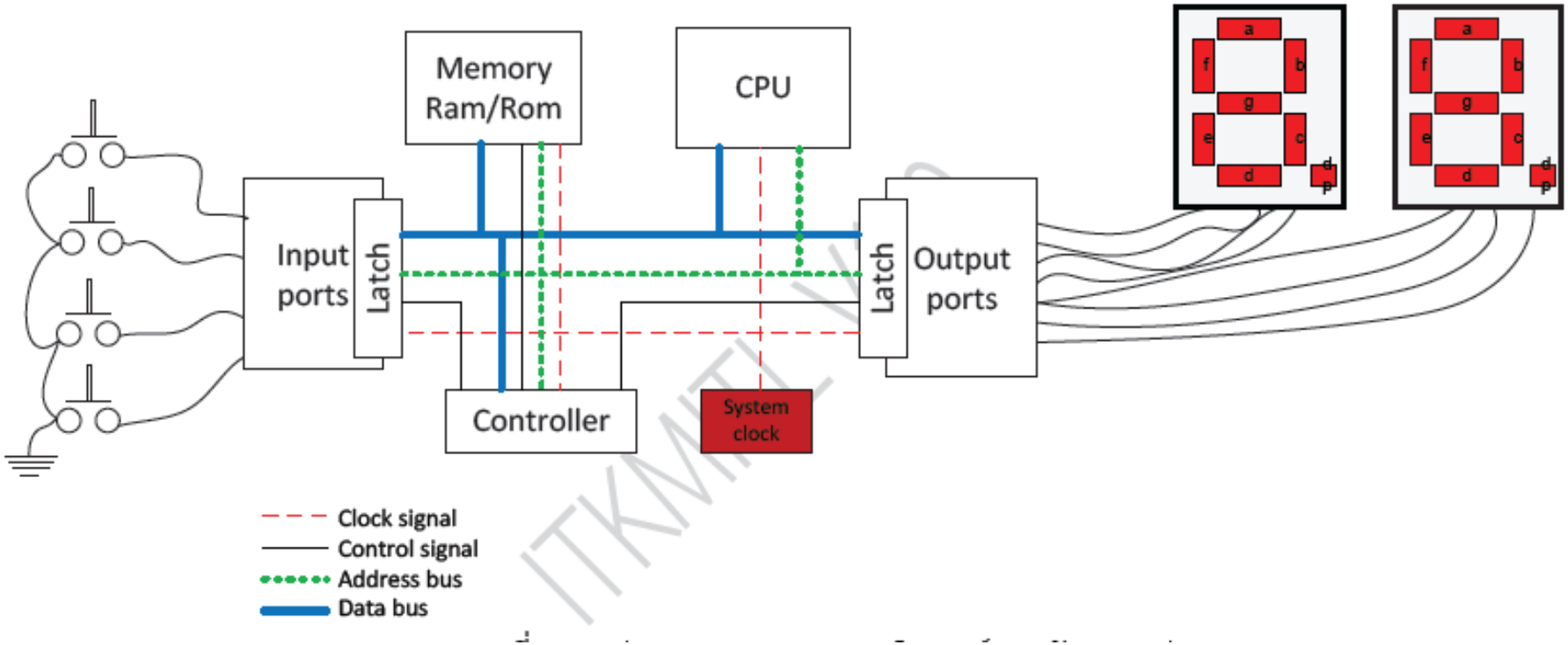
ประมวลผลทั่วไป แต่ไม่เร็ว

- Specific instruction set for input/output controlling
- I/O ports and timers
- Example family chips
  - PIC
  - MCS-51
  - ARM

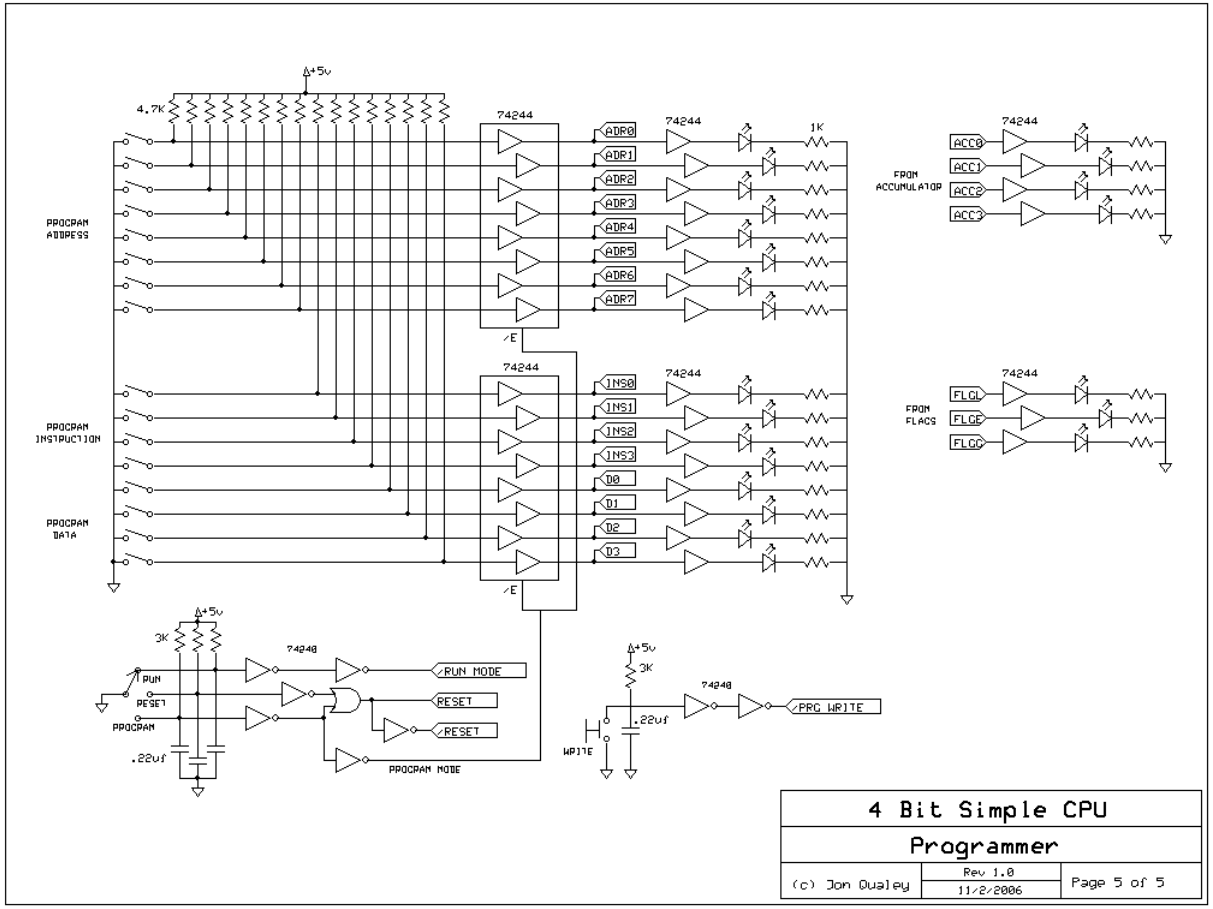
Control เครื่องจักร
. หุ่นยนต์ เครื่องซักผ้า
etc.

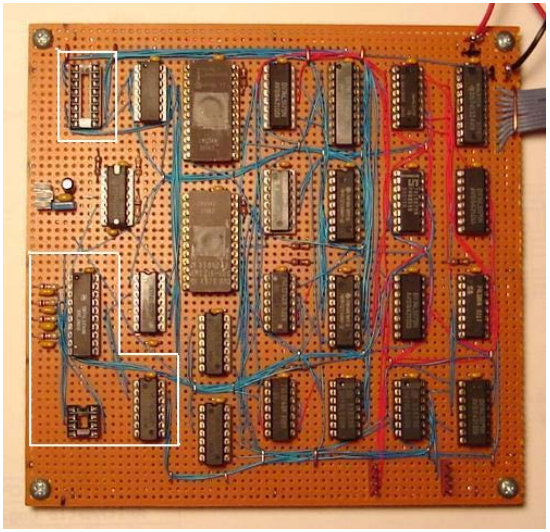


Device Connectivity in the computer



Why does the digital circuit become the computer?





http://www.galacticelectronics.com/4BitCPU_Programmer.HTML

Activity1.4

- Find the CPU specification from the list below with the Internet.
  - Z80 Microprocessor → 1980 → พบในเครื่องเล่นเกม
  - 8051 Microcontroller → เครื่องจักร
  - 68HC11 Microcontroller → เครื่องจักร advanced , รถยนต์ → ระบบในรถคุยกัน ผ่าน Bus
  - PIC 16Fxx Microcontroller → อุปกรณ์อิเลกทรอนิกส์ขนาดเล็ก "มาก"
  - Arduino Microcontroller → Library เยอะ
  - 8088 Microprocessor → 1980 → intel
  - 80286 Microprocessor → intel 16 บิต
  - 80386 Microprocessor → intel 16/32 bit
  - Pentium Microprocessor → รุ่นแรกที่เป็น 64 / สุดท้ายที่เป็น 32

Summary